FOR A CYCLIC TWO-PERIODIC COMPLEX OF A COHERENT SHEAF WITH K-DIMENSIONAL COHOMOLOGY, THE K-CYCLES OF THE TWO COHOMOLOGY SHEAVES ARE RATIONALLY EQUIVALENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02SB (sources/stacks-chow/pinned-original-source.tex line 18417).

Allowed standard prerequisites (editorial).

- The two-periodic complex attached to two endomorphisms of a module and its determinant. Homology, Equation labelled homology-equation-cyclic-complex (Stacks tag 02MW) and Same chapter, Definition labelled definition-periodic-determinant (source line 16611), invoked at statement line 18431 and proof lines 18456-18461. Fix the complex notation (F, phi, psi), its cohomology sheaves, and the determinant of the complex at the generic point of a component.
- Length of modules with finite dimensional support and local finiteness of the top dimensional components. Same chapter, Lemma labelled lemma-length-finite (source line 1702), invoked at proof line 18452. Supplies that the family of components of dimension k+1 is locally finite, so that the sum of divisors is a well defined cycle.
- The length-determinant relation for a two-periodic complex over a one dimensional local ring. Same chapter, Proposition labelled proposition-length-determinant-periodic-complex (source line 18024), invoked at proof line 18505. Supplies that the e-invariant of the localised complex equals the sum of the orders of the localised determinants, which is the coefficient comparison closing the proof; it is retained only as a tagged navigation pointer.
- The base situation (S, delta) and the rational equivalence of k-cycles. Same chapter, Situation labelled situation-setup (source line 1372), invoked in the statement. Fixes delta-dimensions, the attached k-cycles and the relation $\sim$ rat.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 02MW: sources/stacks-chow/tags.txt, source lines 3459–3459. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02QW: sources/stacks-chow/pinned-original-source.tex, source lines 1702–1702. Same-chapter pointer into the author environment lemma at source line 1702; invoked at proof line 18452 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02PJ: sources/stacks-chow/pinned-original-source.tex, source lines 16611–16611. Same-chapter pointer into the author environment definition at source line 16611; invoked at proof line 18461 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 02QD: sources/stacks-chow/pinned-original-source.tex, source lines 18024–18024. Same-chapter pointer into the author environment proposition at source line 18024; invoked at proof line 18505 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $\mathcal{F}$ be a coherent sheaf on X. Let*

$$\ldots \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F} \xrightarrow{\varphi} \mathcal{F} \longrightarrow \ldots$$

be a complex as in Homology, Equation (02MW). Assume that

(1) $\dim_\delta(\mathit{Supp}(\mathcal{F})) \leq k + 1$.
(2) $\dim_\delta(\mathit{Supp}(H^i(\mathcal{F}, \varphi, \psi))) \leq k$ *for* $i = 0, 1$.

Then we have

$$[H^0(\mathcal{F}, \varphi, \psi)]_k \sim_{rat} [H^1(\mathcal{F}, \varphi, \psi)]_k$$

as k-cycles on X.

Proof. Let $\{W_j\}_{j \in J}$ be the collection of irreducible components of $\mathrm{Supp}(\mathcal{F})$ which have δ-dimension $k+1$. Note that $\{W_j\}$ is a locally finite collection of closed subsets of X by Lemma 02QW. For every j, let $\xi_j \in W_j$ be the generic point. Set

$$f_j = \det_{\kappa(\xi_j)}(\mathcal{F}_{\xi_j}, \varphi_{\xi_j}, \psi_{\xi_j}) \in R(W_j)^*.$$

See Definition 02PJ for notation. We claim that

$$-[H^0(\mathcal{F}, \varphi, \psi)]_k + [H^1(\mathcal{F}, \varphi, \psi)]_k = \sum (W_j \to X)_* \mathrm{div}(f_j)$$

If we prove this then the lemma follows.

Let $Z \subset X$ be an integral closed subscheme of δ-dimension k. To prove the equality above it suffices to show that the coefficient n of $[Z]$ in $[H^0(\mathcal{F}, \varphi, \psi)]_k - [H^1(\mathcal{F}, \varphi, \psi)]_k$ is the same as the coefficient m of $[Z]$ in $\sum (W_j \to X)_* \mathrm{div}(f_j)$. Let $\xi \in Z$ be the generic point. Consider the local ring $A = \mathcal{O}_{X,\xi}$. Let $M = \mathcal{F}_\xi$ as an A-module. Denote $\varphi, \psi : M \to M$ the action of φ, ψ on the stalk. By our

choice of $\xi \in Z$ we have $\delta(\xi) = k$ and hence $\dim(\mathrm{Supp}(M)) = 1$. Finally, the integral closed subschemes W_j passing through ξ correspond to the minimal primes $\mathfrak{q}_i$ of $\mathrm{Supp}(M)$. In each case the element $f_j \in R(W_j)^*$ corresponds to the element $\det_{\kappa(\mathfrak{q}_i)}(M_{\mathfrak{q}_i}, \varphi, \psi)$ in $\kappa(\mathfrak{q}_i)^*$. Hence we see that

$$n = -e_A(M, \varphi, \psi)$$

and

$$m = \sum \mathrm{ord}_{A/\mathfrak{q}_i}(\det_{\kappa(\mathfrak{q}_i)}(M_{\mathfrak{q}_i}, \varphi, \psi))$$

Thus the result follows from Proposition 02QD. □